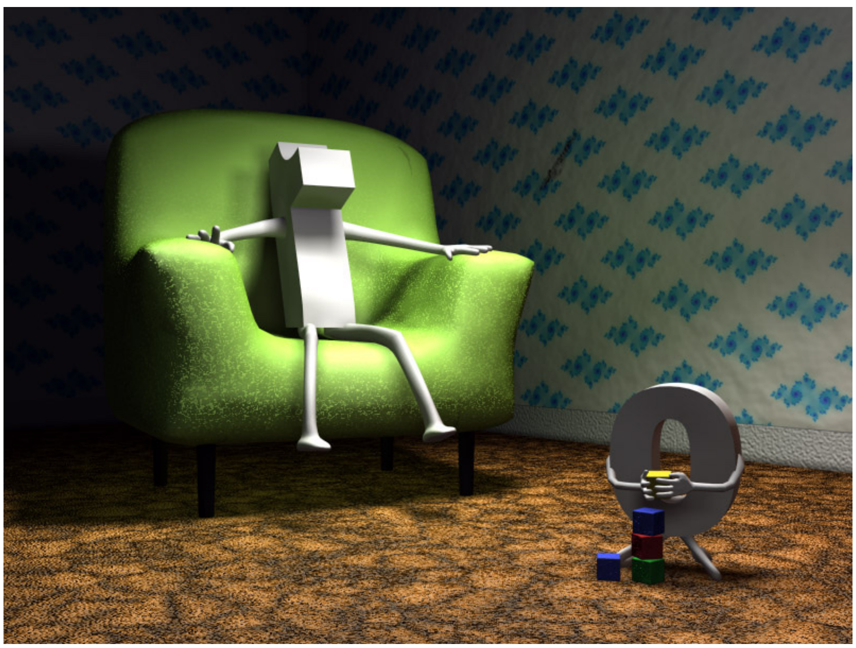
# Chapter 1. INTEGERS

## 1. 2 Divisibility and Prime Factorisation

### 1.2.2 Gcd. Primes and coprime integers



**Today we will prove a very important result:**

**EUCLID's LEMMA**

THEOREM **1.17.** *Let $m$ and $n$ be* **relatively prime** *integers. Then for any integer $k$*

(1) $m|k,\ n|k \Rightarrow mn|k$.

(2) $m|kn \Rightarrow m|k$.

Proof. By Bézout's Lemma we have that $1 = xm + yn$, for some integers $x$ and $y$.

(1) $m|k \Rightarrow k = q_1 m$, for some $q_1 \in \mathbb{Z}$

$n|k \Rightarrow k = q_2 n$, for some $q_2 \in \mathbb{Z}$

$$k = 1 \cdot k = (xm + yn)k = (xm)k + (yn)k =$$

$$= (xm)(q_2 n) + (yn)(q_1 m) = (xq_2 + yq_1)mn$$

$$\Rightarrow mn|k$$

(2) $m|kn \Rightarrow kn = qm$ for some integer $q$.

$$k = 1 \cdot k = (xm + yn)k = (xk)m + y(nk)$$

$$= (xk)m + y(qm) = (xk + yq)m$$

$$\Rightarrow m \mid K \qquad \blacksquare$$

<u>Know fact</u>

$$\left.\begin{array}{l} m, n \text{ integers} \\ mn \text{ even}\end{array}\right\} \Rightarrow \begin{array}{c} m \text{ is even} \\ \text{or} \\ n \text{ is even}\end{array}$$

$$2 \mid mn \Rightarrow 2 \mid m \text{ or } 2 \mid n$$

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**EUCLID'S LEMMA.** Let $p$ be a prime.

(1) If $p|mn$ where $m$ and $n$ are integers, then either $p|m$ or $p|n$.

(2) If $p|m_1m_2\ldots m_r$ where each $m_i$ is an integer, then $p|m_i$ for some $i$.

<u>Proof</u>. Let $d = \gcd(m, p)$. Then by definition $d \geq 1$ and $d \mid p$. From here since $p$ is prime, we have that either $d = 1$ or $d = p$.

If $d = p$, then $p \mid m$ and we are done. Otherwise, if $d = 1$ then $m$ and $p$ are coprime.

$$\left.\begin{array}{l} p \mid mn \\ \\ p,\ m \text{ are coprime} \end{array}\right\} \overset{\text{thm 1.17(2)}}{\Longrightarrow} p \mid n$$

<u>Remark</u>. Euclid's Lemma might not be true if $p$ is <u>NOT</u> a prime..

$$6 \mid 3 \cdot 4 \quad \text{and} \quad 6 \nmid 3 \quad \text{and} \quad 6 \nmid 4.$$